RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie
....
MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR
....
COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN HIGHER NATIONAL DIPLOMA EXAM (HND)
....

REPUBLIC OF CAMEROON
Peace-Work-Fatherland
....
MINISTRY OF HIGHER EDUCATION
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NATIONAL COMMISSION FOR THE ORGANISATION OF HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM
....

National Exam of Higher National Diploma-New program – 2021 Session

Spécialty/option : SWE-NWS-EDM
Paper : Discrete Mathematics
Duration : 4hours

Credits : 4/4/10

Instructions: *Answer all questions. You are authorized to use only non-programmable calculator.*

SECTION A: MCQS (20 MARKS)

1) **The probability of selecting a man at random in a crowd containing 20 men and 33 women is,**
   A. 0.6226 B. 0.05 C. 0.3774 D. 1

2) **The expectation of obtaining a 4 upwards with 3 throws of a fair dice is,**
   A. 0.167 B. 0.750 C. 0.666 D. 0.500

3) **The probability of selecting at random the winning horses in both the first and second races if there are 10 horses in each race**
   A. 0.10 B. 0.20 C. 0.01 D. 0.02

4) **The expectation E is equal to;**
   A. pn B. $\frac{\sum_{i=1}^{n} x_i}{\mu}$ C. $\frac{\sum_{i=1}^{n} x_i}{\sigma}$ D. μx_i

5) **The probability of r successes in n trials in a binomial event is given by**
   A. $\binom{n}{r}q^{n-r}p^{r}$ B. $\binom{r}{n}q^{n-r}p^{n}$ C. $\binom{n}{r}q^{n-1}p^{r}$ D. $\binom{n}{1}q^{n-1}p^{r}$

6) **The probability distribution of tossing a fair coin n times is given by**
   A. np B. $n\sigma$ C. $(p+q)^{n}$ D. $(p+q)^{n+1}$

7) **For the random variables $x_i = \{1,2,3,4,5,6\}$, for the possible outcomes of throwing a fair die, what is the probability $p(x \geq 3)$?**
   A. 2/6 B. 3/6 C. 4/6 D. 1/6

8) **The probability of having 0 head in two tosses of a fair coin is**
   A. ¼ B. ½ C. 1 D. 0

9) The Laplace transform $\mathcal{L}\{sin2t\}$ is

A. $\frac{2}{s^2+2^2}$ B. $\frac{2s}{s^2-2^2}$ C. $\frac{s}{s^2+2^2}$ D. $\frac{s}{s^2-2^2}$

10) The probability of having at least 1 head in two tosses of a fair coin is

A. ¼ B. ½ C. ¾ D. 1

11) The following equations: $\frac{tanx+secx}{secx(1+\frac{tanx}{secx})}$ is equal to:

A. $cosx$ B. $cos3x$ C. 1 D. $sinx$

12) $\lim_{x\to 0}\left\{\frac{tanx-x}{x^3}\right\}$ is equal to

A. 3 B. 1/3 C. 2/3 D. 0

13) The Laplace transform $\mathcal{L}\{2e^{7x-2}\}$ is equal to

A. $\frac{2e^{-2}}{s-7}$ B. $\frac{2e^{2}}{s-7}$ C. $\frac{e^{-2}}{2(s-7)}$ D. $\frac{e^{2}}{2(s-7)}$

14) The general term of the series $1+\frac{3}{2}+\frac{5}{2^2}+\frac{7}{2^3}+\cdots$, is

A. $\frac{2n+1}{2^{n-1}}$ B. $\frac{2n-1}{2^{n+1}}$ C. $\frac{2n-1}{2^{n-1}}$ D. $\frac{2n+1}{2^{n+1}}$

15) *When the* $\lim_{x\to\pm\infty} f(x)=\pm\infty$, it implies Possibility of

A. a vertical asymptote
B. a horizontal asymptote
C. a quadratic asymptote
D. an oblique asymptote

16) The Domain of definition of the function: $f(x)=\frac{x^2-1+\ln x}{e^x-e}$, is

A. $]0,1[\cup]1,+\infty[$
B. $]1,+\infty[$
C. $]0,e[\cup]e,+\infty[$
D. $]e,1[\cup]1,+\infty[$

17) the parity of the function $f(x)=\frac{x^3-x}{|x|+4}$ is,

A. even B. odd C. neither odd nor even D. positive

18) The equation, *log(x -1)+log(x+1)=2log(x+2)* has as solution x=

A. 4/5 B. -5/4 C. 5/4 D. -4/5

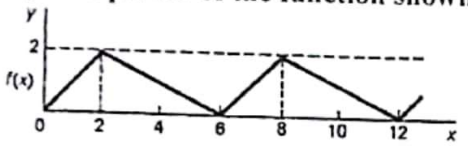
19) In the D'Alambert's theorem the $\lim_{n\to\infty}\frac{U_{n+1}}{U_n} =< 1$ implies

A. Convergence B. Divergence C. Inconclusive D. Constant

20) Three numbers are in arithmetic progression. Their sum is 9 and their product is 20.25. What are the three numbers?

A. 5/2, 1/2, 3/2 B. 7/2, 5/2, 6/2 C. 3/2, 3, 9/2 D. 5/2, 3, ½

21) The period of the function shown graphically below is

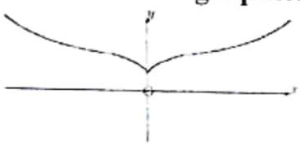


A. 2 B. 8 C. 6 D. 3

22) The exact solution to the following; $\int_{\frac{1}{2}}^{1} \frac{x}{\sqrt{4x^2-1}}\, dx$ is

A. $\sqrt{3}/2$ B. $2\sqrt{3}$ C. $\frac{\sqrt{3}}{8}$ D. $\frac{\sqrt{3}}{4}$

23) The function shown graphically below is



A. an even function
B. an odd function
C. neither odd nor even function
D. an asymmetric function

24) If $\ln(1+y) = \frac{1}{2}x^2 + ln4$ then

A. $y = \frac{1}{2}x^2 + ln4 - 1$ C. $y = 4e^{\frac{1}{2}x^2} - ln4$

B. $y = 4e^{\frac{1}{2}x^2} - 1$ D. $y = \frac{1}{2}x^2 - 4$

25) Find the value(s) of $\frac{dy}{dx}$ of $x^2y + y^2 = 5$ at $y = 1$

A. $-\frac{2}{3}$ *only* B. $\frac{2}{3}$ *only* C. $\pm\frac{2}{3}$ D. $\pm\frac{4}{3}$

26) When $f(t+T) = f(t)$, T is called

A. the fundamental period of $f(t)$ C. constant of $f(t)$
B. the period of $f(t)$ D. increment of $f(t)$

27) Which theorem is verified if $f(x)$ is continuous on $[a,b]$, differentiable on $]a,b[$, $f(a) = f(b)$ and there exist at least one number c within $]a,b[$, such that $f'(c) = 0$.

A. Cauchys theorem

B. Green's theorem

C. Rolle's theorem

D. Mean value theorem

28) The following equation, $\oint Pdx + Qdy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy$, expresses the:

A. Cauchy's theorem

B. Green's theorem

C. Rolle's theorem

D. Mean value theorem

29) $tanh^{-1}\left(\frac{x^2-1}{x^2+1}\right)$ is identical to

A. $tan\left(\frac{x^2-1}{x^2+1}\right)$ B. $tan^{-1}\left(\frac{x^2-1}{x^2+1}\right)$ C. lnx D. $\ln(x^2-1)$- $\ln(x^2+1)$

30) If $f(x) = 3^x$, then $f'(x) =$;

A. $ln3x$ B. $3^x ln3$ C. 3^x D. $3x^2$

31) The value of x in the equation $3.72 = ln\left(\frac{5.14}{x}\right)$ is

A. $x = \frac{5.14}{e^{3.17}}$. B. $x = \frac{e^{3.17}}{5.14}$ C. $x = 5.14e^{3.17}$ D. $x = 3.17e^{5.14}$

32) If $y = \ln(7-4x)$ then $\frac{dy}{dx}$ is

A. $\frac{4}{7-4x}$ B. $\frac{-4}{7+4x}$ C. $\frac{4}{7+4x}$ D. $\frac{-4}{7-4x}$

33) Consider the sequence $U_n = \frac{1}{2n} - \frac{1}{2(n+2)}$. The first term of the series S_1

A. $S_1 = 1/2$ B. $S_1 = 1/6$ C. $S_1 = 1/3$ D. $S_1 = 0$

34) The domain of definition of the function $f(x) = \frac{x}{\sqrt{|x+2|}} - \frac{x}{\sqrt{|x+1|}}$ is :

A. $]-\infty, -2[U]-2, -1[U]-1, +\infty[$

B. $]-2, -1[$

C. $]-2, -1[U[-1, +\infty[$

D. $]-\infty, -2]U]-2, +\infty[$

35) From the Cauchy's criteria, $\lim_{n\to\infty} \sqrt[n]{U_n} < 1$, implies

A. divergence B. inconclusive C. convergence D. constant

36) The value(s) of $x = C$ for $f(x) = x^3 + 2x^2 - x$ on $[-1,2]$, for which the mean value theorem is verified is

A. $C = \frac{-2\pm\sqrt{19}}{3}$ B. $C = \frac{-2-\sqrt{19}}{3}$ *only* C. $C = \frac{2\pm\sqrt{19}}{3}$ D. $C = \frac{-2+\sqrt{19}}{3}$ *only*

37) For three vectors $v_{1,}v_{2,}v_3$ in a vector space and constants $c_{1,}c_{2,}c_{3,}$ from the set of real numbers, $c_1v_1 + c_2v_2 + c_3v_3 = 0$; with c_1, c_2, c_3 not all zero then the set of vectors $v_{1,}v_{2,}v_3$ are said to be

A. linearly dependent B. linearly independent C. a kernel D. an endomorphism

38) The gradient of the curve $y = 3x^4 - 2x^2 + 5x - 2$ at the point $(0, -2)$ is

A. 2 B. 4 C. 5 D. 6

39) If $y = cosh^{-1}x$, then:

A. $coshx = 1/x$ B. $coshx = 1/y$ C. $coshy = y$ D. $coshy = x$

40) $\left(x + \sqrt{x^2 - 1}\right)\left(x - \sqrt{x^2 - 1}\right) =$

A. x^2 B. 1 C. 2 D. $x^2 - 1$

SECTION B: STRUCTURAL (80 MARKS)

1. ANALYSIS (30 Marks)

1.1 (a) A real valued function $f(x)$ is defined by $f(x) = \sqrt{x^2 + x - 12}$. Determine the domain and range of $f(x)$.

(b) (i) Using any of the limit theorems, evaluate $\lim_{n \to \infty} \frac{1+5\cdot10^n}{5+3\cdot10^n}$.

(ii) If $U_{n+1} = \sqrt{U_n + 1}$, $U_1 = 1$, prove that: $\lim_{n \to \infty} U_n = \frac{1}{2}(1 + \sqrt{5})$.

(5 + 5 marks)

1.2 (a) Using the Laplace Transforms or the method of undetermined coefficients, solve the differential equation $Y''(x) + Y(x) = x$, $Y(0) = 0$, $Y'(0) = 2$.

(b) Given the function, $f(x)$, where $f(x) = \begin{cases} 0 , & -5 < x < 0 \\ 3 , & 0 < x < 5 \end{cases}$, $period = 10$,

Determine the Fourier coefficients and write the corresponding Fourier series.**(5+5 marks)**

1.3 Consider the vector field $\vec{F} = (y^2 \cos x + z^3)\vec{\imath} + (2y \sin x - 4)\vec{\jmath} + (3xz^2 - 2)\vec{k}$.

(a) Show that the curl of $\vec{F}$ is 0 i.e. $\vec{\nabla} \times \vec{F} = 0$.

(b) Determine the scalar field ϕ such that $\vec{F} = \vec{\nabla}\phi$.

(c) Determine $\nabla^2\varphi$ if $\varphi(x, y) = e^{\sqrt{x+y}}$. **(3 + 4 + 3 marks)**

2. STATISTICS (30 Marks)

2.1 The table shows the marks, collected into groups, for 400 candidates in an HND examination. The maximum mark was 99.

Marks	No. of Candidates
0 – 9	10
10 – 19	25
20 – 29	45
30 – 39	65
40 – 49	80
50 – 59	70
60 – 69	55
70 – 79	30
80 – 89	15
90 – 99	5

(i) Compile the cumulative frequency table and draw the cumulative frequency curve. Use your curve to estimate:

(ii) Median.

(iii) The $30th$ percentile.

(vi) If the minimum mark for Grade A was fixed at 74, estimate from your curve the percentage of candidates who will obtain Grade A. **(6 + 4 + 4 + 4 marks)**

2.2 The following table summarizes the masses, measured to the nearest microgram (μg), of 200 microchips of the same type.

$Mass\ (\mu g)$	$Frequency$
70 – 79	7
80 – 84	30
85 – 89	66
90 – 94	57
95 – 99	27
100 – 109	13

(i) Calculate estimates of the median and upper quartile of the distribution.

(ii) Estimate the number of microchips whose actual masses are less than 81 μg.

(iii) Calculate estimates of the mass and the standard deviation of the distribution. **(5 + 2 + 5 marks)**

3. PROBABILITY (20 Marks)

3.1 An electronic assembly firm buys its microchips from three different suppliers; half of them are bought from firm X, whilst firms Y and Z supply 30 % and 20 % respectively. The suppliers use different quality – control procedures and the percentages of defective chips are 2 %, 4 % and 4 % for X, Y and Z respectively. The probabilities that a defective chip fail two or more assembly – line test are 40 %, 60 % and 80 %, respectively, whilst all defective chips have a 10 % chance of escaping detection. An assembler finds a chip that fails only one test. What is the probability that it came from supplier X? **(10 marks)**

3.2 The number of errors needing correction on each page of a set of proofs follows a Poisson's distribution of mean μ. The cost of the first correction on any page is α and that of each subsequent correction on the same page is β. Prove that the average cost of correction on a page is $\alpha + \beta(\mu - 1) - (\alpha - \beta)e^{-\mu}$. **(10 marks)**